

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51 **COURSE NAME:** Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

Comparative Analysis

Code of Conduct:

I Pusalapati Chandu(192372119) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Annexure A

Comparative Analysis

1. Apply **SSL/TLS and IPsec protocols** to a secure communication scenario and determine which provides better security for web-based and network-layer protection.
2. Design a secure email system by comparing **end-to-end encryption and gateway-based encryption**, and justify the most suitable approach for enterprise use.
3. Apply **HMAC and Digital Signature techniques** to a data transmission scenario and analyze which method provides better authentication and integrity.
4. Create a **spam detection system** by comparing supervised learning and unsupervised learning approaches, and evaluate their effectiveness in real-world email filtering.
5. Design a secure communication framework by comparing **AES and RSA encryption techniques**, and determine their roles in achieving confidentiality and efficiency.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

QUESTION 1 – SSL/TLS AND IPsec COMPARATIVE ANALYSIS

Question: Apply SSL/TLS and IPsec protocols to a secure communication scenario and determine which provides better security for web-based and network-layer protection.

□filecite□turn2file0□L20-L22□

1. Introduction

SSL/TLS and IPsec are both security technologies, but they operate at different layers of network communication. TLS is mainly used to secure application traffic such as HTTPS. IPsec works at the IP/network layer and can protect IP packets between hosts, gateways or networks.

Therefore, it is not correct to say that one protocol is always better than the other. The correct choice depends on what needs to be protected: a web application connection or network-layer traffic.

2. Secure Communication Scenario

- A user opens a banking website using HTTPS.
- The browser establishes a TLS connection with the web server.
- The server proves its identity using a digital certificate.
- Application data is encrypted and authenticated during transmission.
- For an organization connecting two office networks through an untrusted Internet connection, IPsec can create a secure VPN tunnel.
- The VPN protects IP traffic between the two networks.

3. SSL/TLS – How It Works

1. The client starts communication with a TLS handshake.
2. The server sends its certificate and establishes the cryptographic parameters.
3. The client validates the server identity through the certificate trust chain.
4. The connection establishes symmetric traffic keys.
5. HTTP data is encrypted and authenticated using the established secure session.
6. The protected data travels between the browser and web server.

4. IPsec – How It Works

7. The communicating devices establish an IPsec security association.
8. Cryptographic keys and security parameters are established.
9. IP packets are protected using IPsec mechanisms such as ESP.
10. In transport mode, the payload is protected while the original IP header remains available.
11. In tunnel mode, the original IP packet is encapsulated inside a new packet.
12. The receiving IPsec endpoint verifies and decrypts the protected traffic.

5. Main Comparison

Aspect	SSL/TLS	IPsec
Primary layer	Transport/session and above in practical use	Network/IP layer
Typical use	HTTPS, secure web services, APIs	VPNs and network-to-network protection

Protection scope	Application communication/session	IP packets
Authentication	Certificates and cryptographic handshake	Security associations and configured authentication
Encryption	Protects application data	Protects IP traffic according to mode
Best example	Browser ↔ web server	Office ↔ office VPN
Deployment	Usually application/service based	Usually host, router or gateway based

6. Security Analysis

- TLS is very suitable for web-based security because it protects the actual web session between the browser and server.
- TLS allows a user to verify the identity of the server using certificates.
- IPsec can protect more than one application because it works at the IP layer.
- An IPsec VPN can protect many applications without configuring each application separately.
- IPsec tunnel mode is particularly useful when complete internal packets must cross an untrusted network.
- TLS does not automatically protect unrelated applications that are not using TLS.
- IPsec does not replace application-level authentication and secure application design.

7. Which Provides Better Security?

For web-based protection, SSL/TLS is the more suitable choice because HTTPS directly protects browser-to-web-server communication. It provides server authentication, confidentiality and integrity for the application session.

For network-layer protection, IPsec is more suitable because it protects IP packets and can create VPNs between networks or gateways.

Therefore, the best answer is not that TLS or IPsec is universally stronger. TLS is better suited to web/application security, while IPsec is better suited to network-layer and VPN security.

8. Justification

- Use TLS when the main requirement is secure access to websites, web applications or APIs.
- Use IPsec when the requirement is to secure IP communication across a network boundary.
- Use both when an organization wants network-layer protection plus application-layer security.
- Layered security is stronger because a failure in one security layer does not automatically expose every other layer.

9. Conclusion

SSL/TLS and IPsec solve different security problems. TLS is the preferred technology for web-based communication such as HTTPS, while IPsec is appropriate for network-layer protection and VPN communication. The correct selection should therefore be based on the security scope and communication environment.

QUESTION 2 – END-TO-END VS GATEWAY-BASED EMAIL ENCRYPTION

Question: Design a secure email system by comparing end-to-end encryption and gateway-based encryption, and justify the most suitable approach for enterprise use. □filecite□turn2file0□L23-L23□

1. Introduction

Enterprise email contains confidential information such as business documents, customer information and internal communication. Encryption can protect this information while it travels through networks. Two important approaches are end-to-end encryption and gateway-based encryption.

End-to-end encryption protects the message from the sender to the intended recipient. Gateway-based encryption places encryption and decryption responsibility at an organizational email gateway.

2. End-to-End Encryption

- The sender encrypts the email using the recipient's public key or an equivalent secure mechanism.
- The encrypted message travels through mail servers and networks.
- Intermediate mail servers cannot normally read the protected message content.
- The recipient uses the corresponding private key to decrypt the message.
- Digital signatures can be added to provide authentication and integrity.

3. Gateway-Based Encryption

- The organization operates a secure email gateway.
- The gateway can inspect, classify and encrypt outgoing messages according to company policy.
- The gateway can decrypt or process incoming messages before delivery to internal users.
- Security policies can be applied centrally.
- This approach is easier to manage when an organization needs centralized control over many users.

4. Comparison

Aspect	End-to-End Encryption	Gateway-Based Encryption
Protection	Message protected from sender to recipient	Gateway protects/enforces security at organizational boundary
Visibility to mail infrastructure	Much less plaintext exposure	Gateway may access plaintext to perform policy/security functions
Central control	Lower	High
User key management	More important	Can be centralized
DLP / inspection	More difficult because content is encrypted	Easier at gateway
Enterprise policy enforcement	More complex	Strong centralized control
Privacy from intermediaries	Strong	Depends on gateway trust

Deployment	Can be more complex for users	Often easier to administer centrally
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5. Secure Enterprise Design

13. Create a centralized enterprise identity and certificate/key-management system.
14. Use strong authentication for employees.
15. Use end-to-end encryption for highly sensitive messages when confidentiality from intermediate systems is required.
16. Deploy secure email gateways for malware scanning, spam filtering, DLP and policy enforcement.
17. Protect gateway private keys using strong access controls.
18. Maintain logging and monitoring without unnecessarily exposing message content.
19. Define policies for key recovery, revocation, employee departure and lost credentials.

6. Advantages of End-to-End Encryption

- Provides strong confidentiality between the communicating endpoints.
- Reduces the number of systems that need access to plaintext.
- Useful for highly sensitive information.
- Can support sender authentication and message integrity through digital signatures.

7. Advantages of Gateway-Based Encryption

- Centralized administration is easier for large organizations.
- Security policies can be applied consistently.
- The gateway can integrate with spam filtering, malware scanning and DLP.
- Key and certificate operations can be managed centrally.
- Users do not need to understand every cryptographic operation.

8. Disadvantages

Approach	Main limitation
End-to-End	Key management, user experience and security-policy inspection are more difficult.
Gateway-Based	The gateway becomes a trusted security point and may have access to plaintext.

9. Which Is Most Suitable?

For a large enterprise, gateway-based encryption is often easier to operate because security policies, monitoring, malware controls and encryption can be centralized. However, for highly confidential communication where even the organization's gateway should not read the message, end-to-end encryption is stronger.

A practical enterprise design can therefore use a hybrid approach: gateway security for normal enterprise mail and end-to-end encryption for specially classified or highly sensitive messages.

10. Conclusion

End-to-end encryption provides stronger endpoint-to-endpoint confidentiality, while gateway-based encryption provides better centralized enterprise management. A hybrid design gives an organization both operational control and stronger protection for highly sensitive messages.

QUESTION 3 – HMAC VS DIGITAL SIGNATURE

Question: Apply HMAC and Digital Signature techniques to a data transmission scenario and analyze which method provides better authentication and integrity. □ filecite □ turn2file0 □ L24-L25 □

1. Introduction

HMAC and digital signatures can both help detect data modification and authenticate the source of a message, but they use different cryptographic models. HMAC uses a shared secret key, while a digital signature uses a private/public key pair.

2. Example Scenario

Suppose a company sends an important transaction message from a server to another server. The receiver must determine whether the message was modified and whether it came from an authorized sender.

- Message: Transfer amount = ₹50,000.
- The sender calculates an authentication value.
- The receiver verifies the received data.
- If verification fails, the receiver rejects the message.

3. HMAC Process

20. Sender and receiver share a secret key.
21. Sender calculates HMAC over the message using the secret key.
22. Sender sends the message and HMAC.
23. Receiver uses the same secret key to calculate a new HMAC.
24. Receiver compares the calculated HMAC with the received HMAC.
25. If they match, the message is considered authentic and unchanged under the shared-key assumption.

4. Digital Signature Process

26. Sender generates a hash of the message.
27. Sender signs the hash using the sender's private key.
28. Sender sends the message and digital signature.
29. Receiver calculates the hash of the received message.
30. Receiver verifies the signature using the sender's public key.
31. If verification succeeds, the receiver has evidence that the signature corresponds to the sender's private key and the signed data was not modified.

5. Comparison

Aspect	HMAC	Digital Signature
Key type	Shared secret key	Private/public key pair
Speed	Generally fast	Generally more computationally expensive
Integrity	Yes	Yes
Authentication	Shared-key authentication	Public-key based authentication
Non-repudiation	No strong non-repudiation	Can provide stronger non-repudiation properties
Key distribution	Secret must be shared securely	Public key can be distributed
Best use	Trusted systems sharing a secret	Multi-party authentication and signed documents

6. Which Provides Better Authentication?

For authentication involving many independent parties, digital signatures are generally more suitable because the sender can keep the private key secret while distributing the public key for verification. With HMAC, both parties possess the same secret, so the receiver cannot independently prove to a third party which party created the HMAC.

Therefore, digital signatures provide stronger evidence of origin in situations where independent verification is required.

7. Which Provides Better Integrity?

Both HMAC and digital signatures can provide strong integrity protection when implemented correctly. Any modification to the message should cause verification to fail.

- HMAC is usually faster and efficient for systems that already share a secure secret.
- Digital signatures are useful when verification must be possible using a public key.
- The choice depends on key-management requirements and whether non-repudiation is needed.

8. Practical Recommendation

Situation	Recommended method	Reason
Internal service-to-service communication with shared secret	HMAC	Fast and efficient.
Signed contracts or official documents	Digital signature	Provides public verification.
Communication among many independent organizations	Digital signature	Avoids one shared secret between all parties.
High-throughput trusted backend systems	HMAC	Low computational overhead.

9. Conclusion

HMAC and digital signatures both provide integrity and authentication, but they solve different key-management problems. HMAC is faster and suitable when trusted parties already share a secret. Digital signatures are more suitable when public verification, scalable authentication and stronger non-repudiation properties are required.

QUESTION 4 – SUPERVISED VS UNSUPERVISED SPAM DETECTION

Question: Create a spam detection system by comparing supervised learning and unsupervised learning approaches, and evaluate their effectiveness in real-world email filtering.

□filecite□turn2file0□L26-L28□

1. Introduction

Spam detection can use machine-learning techniques to identify unwanted or malicious email. Two possible approaches are supervised learning and unsupervised learning. The main difference is whether the training data already contains correct labels.

2. Supervised Learning

- Uses labelled email data.
- Each training email is marked as spam or legitimate.
- The algorithm learns patterns associated with each class.
- A new email is classified using the learned model.
- Examples include Logistic Regression, Naive Bayes, Decision Trees and Support Vector Machines.

3. Supervised Spam Detection – Process

32. Collect a large email dataset.
33. Label each message as spam or legitimate.
34. Extract features such as words, URLs, sender information and message structure.
35. Split the dataset into training and testing data.
36. Train the classifier.
37. Test the classifier on unseen messages.
38. Measure accuracy, precision, recall and F1-score.
39. Deploy and periodically retrain the model with new examples.

4. Unsupervised Learning

- Does not require predefined spam labels.
- The algorithm searches for patterns or groups in the data.
- Clustering can group similar email messages.
- Outlier detection can identify unusual messages.
- It can help discover new spam patterns that were not included in a labelled training set.

5. Comparison

Aspect	Supervised	Unsupervised
Training labels	Required	Not required
Main task	Predict known classes	Discover patterns/groups
Accuracy for known spam	Often high with good data	Can be less predictable
New spam patterns	May require retraining	Can discover unusual patterns
Data preparation	Requires labelled dataset	Less dependence on labels
Evaluation	Straightforward using labelled test data	More difficult because true labels may be unavailable

Enterprise use	Strong primary classifier	Useful as a complementary detection method
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6. Example Features

- Frequency of suspicious words.
- Number and type of URLs.
- Use of URL shorteners.
- Sender/domain characteristics.
- Attachment information.
- Unusual message formatting.
- Repeated messages across many accounts.
- Language patterns associated with phishing or spam.

7. Effectiveness in Real-World Filtering

Supervised learning is usually easier to evaluate because labelled test data allows the organization to measure false positives and false negatives. It can achieve strong performance when the training data represents current spam patterns.

However, attackers continuously change their messages. A model trained only on old examples may fail to identify new campaigns. Unsupervised learning can help discover unusual patterns, but a cluster is not automatically spam and may require human or rule-based interpretation.

8. Best Approach

A hybrid system is usually more practical than using only one approach. A supervised classifier can perform the main spam/legitimate classification, while unsupervised anomaly detection can identify unusual new patterns. Rules, reputation systems and security scanning can provide additional protection.

9. Evaluation Metrics

Metric	Meaning
Accuracy	Overall percentage of correct predictions.
Precision	How many messages predicted as spam are actually spam.
Recall	How many actual spam messages are detected.
F1-score	Balances precision and recall.

For email filtering, precision is important because too many false positives can cause legitimate emails to be blocked. Recall is also important because missed phishing or malicious messages can create security risks.

10. Conclusion

Supervised learning is suitable for reliable classification when labelled training data is available. Unsupervised learning is useful for discovering new patterns and anomalies. A hybrid approach combining supervised classification, anomaly detection and security rules is a strong practical design for real-world email filtering.

QUESTION 5 – AES VS RSA IN A SECURE COMMUNICATION FRAMEWORK

Question: Design a secure communication framework by comparing AES and RSA encryption techniques, and determine their roles in achieving confidentiality and efficiency.

□filecite□turn2file0□L29-L31□

1. Introduction

AES and RSA are both important cryptographic algorithms, but they are designed for different purposes. AES is a symmetric encryption algorithm using the same secret key for encryption and decryption. RSA is an asymmetric cryptographic algorithm using a public key and a private key.

A secure and efficient communication system normally uses both types together instead of using RSA to encrypt large amounts of data.

2. AES

- AES uses a symmetric secret key.
- The same secret key is used for encryption and decryption.
- AES is efficient for large amounts of data.
- It is suitable for files, network traffic and message contents.
- The secret key must be protected from unauthorized users.

3. RSA

- RSA uses a public key and a private key.
- The public key can be distributed to other users.
- The private key must remain secret.
- RSA can be used for protecting small secret values and for digital signatures.
- RSA is computationally more expensive than AES for large data encryption.

4. Comparison

Aspect	AES	RSA
Type	Symmetric	Asymmetric
Keys	One shared secret key	Public + private key
Speed	Fast	Slower for large data
Large data encryption	Highly suitable	Not efficient
Key distribution	Secret key must be shared securely	Public key can be distributed
Digital signature	Not the normal role	Can support digital signatures
Main role	Bulk data encryption	Key protection / signatures

5. Secure Communication Framework

40. The sender obtains the receiver's trusted RSA public key.
41. The sender generates a random AES session key.
42. The sender encrypts the actual message or file using AES.
43. The sender encrypts or protects the AES session key using the receiver's public-key mechanism.
44. The encrypted message and protected session key are transmitted.

45. The receiver uses the private key to recover the AES session key.
46. The receiver uses AES to decrypt the actual message.
47. Authentication and integrity can additionally be provided using a digital signature or authenticated encryption.

6. Why AES Is Used for Data

- AES is fast.
- It is suitable for large files and continuous network traffic.
- It requires less computational effort than public-key encryption for bulk data.
- It makes the overall system efficient.

7. Why RSA Is Used for Key Management

- RSA solves the problem of securely distributing a secret value without directly sharing the private key.
- The receiver's public key can be shared openly.
- Only the receiver's private key should recover protected key material.
- RSA can also support digital signatures for authentication and integrity.

8. Hybrid Encryption – Best Design

The best practical design is hybrid encryption. AES protects the actual message because it is efficient. RSA protects the AES session key because public-key cryptography is useful for key establishment and management.

Stage	Algorithm	Reason
Generate session key	AES key generation	Fresh secret for the data.
Encrypt message	AES	Fast bulk encryption.
Protect session key	RSA	Public-key based key protection.
Authenticate sender	RSA digital signature if required	Supports authentication and integrity.
Decrypt session key	RSA private key	Only intended receiver should recover it.
Decrypt message	AES	Efficient recovery of large data.

9. Security and Efficiency Analysis

Using only AES would create a key-distribution problem because the sender and receiver need a secure way to share the secret key. Using RSA for the complete large message would be inefficient. Hybrid encryption solves both problems by using each algorithm where it is strongest.

- Confidentiality: AES encrypts the message.
- Key protection: RSA protects the session key.
- Authentication: A digital signature can authenticate the sender.
- Integrity: Authenticated encryption or a digital signature can detect modification.
- Efficiency: AES handles the large data while RSA handles only small key material.

10. Conclusion

AES and RSA should not be treated as competing algorithms where one replaces the other. AES is best for efficient bulk-data confidentiality, while RSA is useful for public-key key management and digital signatures. A hybrid AES + RSA framework therefore provides a practical balance between confidentiality, authentication and performance.

FINAL COMPARATIVE SUMMARY

Question	Comparison	Best Practical Choice
Q1	TLS vs IPsec	TLS for web/application protection; IPsec for network/VPN protection.
Q2	End-to-end vs gateway email encryption	Hybrid enterprise model; gateway for centralized control and E2E for highly sensitive content.
Q3	HMAC vs digital signature	HMAC for fast shared-secret systems; digital signature for scalable public verification.
Q4	Supervised vs unsupervised spam detection	Hybrid system using supervised classification plus anomaly detection.
Q5	AES vs RSA	Hybrid encryption: AES for data, RSA for key protection/signatures.

FINAL CONCLUSION

The five comparative questions demonstrate that cybersecurity decisions depend on the communication layer, security requirement, key-management model and performance requirement. A good security solution does not simply choose the algorithm with the highest security level. It chooses the technology that fits the problem.

For web security, TLS is appropriate; for network-layer VPN protection, IPsec is appropriate. For enterprise email, centralized gateway controls can be combined with end-to-end protection for highly sensitive messages. HMAC is efficient for shared-secret authentication, while digital signatures are better when public verification is required. Supervised learning provides strong labelled classification, while unsupervised learning can discover unusual patterns. Finally, AES and RSA work best together in a hybrid design.